



Fr. C. Rodrigues Institute of Technology, Vashi

(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

Department of Computer Engineering

Sem/Div : VII/A

Department of Computer Engineering

Time: 1 Hour

Subject : Natural Language Processing Assignment Test -I (SH-2025)

Total Marks: 20

SET A

Q.	All questions are compulsory.(Each question carries 05 marks)	CO	PO	PSO	BTL
a)	Consider the following corpus • <s> Martin/N Justin/N can/M watch/V Will/N </s> • Spot/N will/M watch/V Martin/N </s> • Will/M Justin/N spot/V Martin/N </s> • Martin/N will/M pat/V Spot/N </s> N: Noun, M: Modal verb, V: Verb Compute the emission and transition probabilities for a bigram HMM.	3	1,2,5,15	1,2	3
b)	Compute the perplexity of the sentence “We enjoy music” using a bigram language model constructed from the given text corpus: We enjoy sports.They play music.We love dancing.Students enjoy learning.Music makes life better.	2	1,2,5,15	1,2	3
c)	Consider the grammar S → NP VP NP → Det N VP → V NP V → includes Det → The Det → a N → meal N → flight Compute the CKY table to check the string “The flight includes a meal”, belongs to the given grammar.	3	1,2,5,15	1,2	3
d)	List and define the types of ambiguities in language processing.	1	1,2, 12	1,2	1